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630.

ON AN EXPRESSION FOR $1 \pm \sin(2p + 1)u$ IN TERMS OF $\sin u$.

[From the *Messenger of Mathematics*, vol. v. (1876), pp. 7, 8.]

Write $\sin u = x$, then we have

$$\begin{array}{ll} \sin u = x, & \cos u = \sqrt{(1 - x^2)}, \\ \sin 3u = 3x - 4x^3, & \cos 3u = (1 - 4x^2)\sqrt{(1 - x^2)}, \\ \sin 5u = 5x - 20x^3 + 16x^5, & \cos 5u = (1 - 12x^2 + 16x^4)\sqrt{(1 - x^2)}, \\ \&c. & \&c. \end{array}$$

It is hence clear, that in general

$$\begin{aligned} 1 - \sin(2p + 1)u &= (1 \pm x)\{(1, x)^p\}^2, \\ 1 + \sin(2p + 1)u &= (1 \mp x)\{(1, -x)^p\}^2, \end{aligned}$$

where $(1, x)^p$ denotes a rational and integral function of x of the order p , and $(1, -x)^p$ the same function of $-x$; for it is only in this manner that we can have

$$\cos^2(2p + 1)u = (1 - x^2) \{(1, x)^p\}^2.$$

We, in fact, find

$$\begin{aligned} 1 + \sin u &= 1 + x, \\ 1 - \sin 3u &= (1 + x)(1 - 2x)^2, \\ 1 + \sin 5u &= (1 + x)(1 + 2x - 4x^2)^2, \\ 1 - \sin 7u &= (1 + x)(1 - 4x - 4x^2 + 8x^3)^2, \\ &\text{\&c.} \end{aligned}$$

and it thus appears that the form is

$$1 + (-)^p \sin(2p + 1)u = (1 + x) \{(1, x)^p\}^2.$$

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To find herein the expression of the factor $(1, x)^p$, write $u = \frac{1}{2}\pi - \theta$ and consequently $x = \cos \theta$; we have therefore

$$1 + \cos(2p+1)\theta = (1 + \cos \theta) \{(1, x)^p\}^2,$$

where in the second factor on the right-hand side x is retained to stand for its value $\cos \theta$. This gives

$$2 \cos^2(p + \frac{1}{2})\theta = 2 \cos^2 \frac{1}{2}\theta \cdot \{(1, x)^p\}^2,$$

or, what is the same thing,

$$(1, x)^p = \frac{\cos(p + \frac{1}{2})\theta}{\cos \frac{1}{2}\theta},$$

viz. this is

$$= \cos p\theta - \sin p\theta \cdot \frac{\sin \frac{1}{2}\theta}{\cos \frac{1}{2}\theta},$$

which is

$$= \cos p\theta - \sin p\theta \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}.$$

We have

$$\cos p\theta + i \sin p\theta = \{x + i\sqrt{(1-x^2)}\}^p = X + i\sqrt{(1-x^2)} Y, \text{ suppose,}$$

where X, Y are rational and integral functions of x of the orders p and $p-1$ respectively; that is,

$$\cos p\theta = X, \quad \sin p\theta = \sin \theta \cdot Y,$$

and we have therefore

$$(1, x)^p = X - Y(1-x),$$

which is the required expression for $(1, x)^p$. For instance
 $p = 3$,

$$X + i\sqrt{(1-x^2)} Y = \{x + i\sqrt{(1-x^2)}\}^3;$$

that is,

$$X = -3x + 4x^3, \quad Y = -1 + 4x^2,$$

and $\therefore -(1-x)Y = 1 - x - 4x^2 + 4x^3$, so that

$$X - (1-x)Y = 1 - 4x - 4x^2 + 8x^3 = (1, x)^3,$$

and hence

$$1 - \sin 7u = (1+x)(1-4x-4x^2+8x^3)^2,$$

which agrees with a result already obtained.

The foregoing value of $(1, x)^p$ may also be written

$$(1, x)^p = \frac{1}{x+1} \left\{ \left(\frac{x+1}{2} \right)^{p+1} - \left(\frac{x-1}{2} \right)^{p+1} \right\},$$

which however is not practically so convenient.

The formula corresponds to a like formula in elliptic functions, viz. writing $\operatorname{sn} u = x$, the numerator of $1 + (-)^p \operatorname{sn}(2p+1)u$ is

$$= (1+x) \{(1, x)^p\}^2,$$

which is $(1+x)$ multiplied by the square of a rational and integral function of x .

631.

SYNOPSIS OF THE THEORY OF EQUATIONS.

[From the *Messenger of Mathematics*, vol. v. (1876),
pp. 39—49.]

The following was proposed, as one of the subjects of a Dissertation for the Trinity Fellowships:

Synopsis of the theory of equations; i.e. a statement in a logical order, of the divisions of the subject and the leading questions and theorems, but without demonstrations.

In the subject “Theory of Equations,” the term *equation* is used to denote an equation of the form $x^n - p_1x^{n-1} + \dots + p_n = 0$, where $p_1, p_2, \dots, p_n$ are regarded as known, and x as a quantity to be determined; for shortness, the equation is written $f(x) = 0$.

The equation may be *numerical*; that is, the coefficients $p_1, p_2, \dots, p_n$ are then numbers; understanding by number, a quantity of the form $\alpha + \beta i$, where α and β have any positive or negative real values whatever; or say, each of these is regarded as susceptible of continuous variation from an indefinitely large negative to an indefinitely large positive value: and i denotes $\sqrt{-1}$.

Or the equation may be *algebraic*; viz. the coefficients are then not restricted to denote, or are not explicitly considered as denoting, numbers.

I. We consider first numerical equations.

A number a (real or imaginary), such that substituted for x it makes the function $x^n - p_1x^{n-1} + \dots \pm p_n$ to be $= 0$, or say, such that it satisfies the equation, is said to be a *root* of the equation; viz. a being a root, we have

$$a^n - p_1a^{n-1} + \dots \pm p_n = 0, \quad \text{or say } f(a) = 0;$$

and it is then shown that $x - a$ is a factor of the function $f(x)$, viz. that we have

$$f(x) = (x - a) f_1(x),$$

where $f_1(x)$ is a function $x^{n-1} - q_1x^{n-2} + \dots \pm q_{n-1}$, of the order $n - 1$, with numerical coefficients $q_1, q_2, \dots, q_{n-1}$.

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In general, a is *not* a root of the equation $f_1(x) = 0$; but it may be so, viz. $f_1(x)$ may contain the factor $x - a$; when this is so, $f(x)$ will contain the factor $(x - a)^2$; writing then $f(x) = (x - a)^2 f_2(x)$, and assuming that a is not a root of the equation $f_2(x) = 0$, $x = a$ is then said to be a *double root* of the equation. Similarly, $f(x)$ may contain the factor $(x - a)^3$ and no higher power, and then $x = a$ is said to be a *triple root*; and so on.

Supposing, in general, that $f(x) = (x - a)^\alpha F(x)$, where α is a positive integer which may be $= 1$, and $F(x)$ is of the order $n - \alpha$, then if b is a root different from a , we shall have $x - b$ a factor (in general a simple one, but it may be a multiple one) of $F(x)$, and $f(x)$ will in this case become

$$f(x) = (x - a)^\alpha (x - b)^\beta \Phi(x),$$

where β is a positive integer which may be $= 1$, and $\Phi(x)$ is of the order $n - \alpha - \beta$. The original equation $f(x) = 0$ is in this case said to have α roots each $= a$, β roots each $= b$, and so on.

We have the theorem, a numerical equation of the order n has in every case n roots, viz. there exist n numbers $a, b, \dots$ (in general, all of them distinct, but they may arrange themselves in groups of equal values) such that

$$f(x) = (x - a)(x - b)(x - c) \dots \text{ identically.}$$

If an equation has equal roots, these can in general be determined; the case is at any rate a special one, which may be here omitted from consideration. It is therefore, in general, assumed that the equation $f(x) = 0$ under consideration has all its roots unequal.

If the coefficients $p_1, p_2, \dots$ are all or any one or more of them imaginary, then the equation $f(x) = 0$, separating the real and imaginary parts, may be written $F(x) + i\Psi(x) = 0$, where $F(x)$, $\Psi(x)$ are each of them a function with real coefficients; and it thus appears that the equation $f(x) = 0$ with imaginary coefficients has not in general any real root; supposing it to have a

real root a , this must be at once a root of each of the equations $F(x) = 0$ and $\Psi(x) = 0$.

But an equation with real coefficients may have as well imaginary as real roots; and we have further the theorem that for such an equation the imaginary roots enter in pairs, viz. $\alpha + \beta i$ being a root, then will also $\alpha - \beta i$ be a root.

Considering an equation with real coefficients, the question arises as to the *number* and *situation* of its real roots; this is completely resolved by means of *Sturm's theorem*, viz. we form a series of functions $f(x)$, $f'(x)$, $f_1(x)$, $\dots$, $f_n(x)$ (a constant) of the degrees n , $n - 1$, $\dots$, 2 , 1 , respectively; and substituting therein for x any two real values a and b , we find by means of the resulting signs of these functions how many real roots of $f(x)$ lie between the limits a , b .

The same thing can frequently be effected with greater facility by other means, but the only general method is the one just referred to.

In the general case of an equation with imaginary (it may be real) coefficients, the like question arises as to the *situation* of the (real or imaginary) roots, viz. if for facility of conception we regard the constituents α , β of a root $\alpha + \beta i$ as the coordinates of a point *in plano*, and accordingly represent the root by such point; then drawing in the plane any closed curve or "contour," the question is how many roots lie within such contour.

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This is solved theoretically by means of a theorem of Cauchy's, viz. writing in the original equation $x + iy$ in place of x , the function $f(x + iy)$ becomes $= P + iQ$, where P and Q are each of them a rational and integral function (with real coefficients) of (x, y) . Imagining the point (x, y) to travel along the contour, and considering the number of changes of sign from $-$ to $+$ and from $+$ to $-$ of the fraction $\frac{Q}{P}$ corresponding to passages of the fraction through zero (that is, to values for which P becomes $= 0$, disregarding those for which Q becomes $= 0$), the difference of these numbers determines the number of roots within the contour. The investigation leads to a proof of the before-mentioned theorem, that a numerical equation of the order n has precisely n roots.

But, for the actual determination, it is necessary to consider a rectangular contour, and to apply to each of its sides separately a method such as that of Sturm's theorem; and thus the actual determination ultimately depends on a method such as that of Sturm's theorem.

Recurring to the case of an equation with real coefficients, it is important to separate the real roots, viz. to determine limits, such that each real root lies alone by itself between two limits l and m . This can be done (with more or less difficulty according to the nearness of the real roots) by repeated applications of Sturm's theorem, or otherwise.

The same thing would be useful, and can theoretically be effected, in regard to the roots of an equation generally, viz. we may, by lines parallel to the axes of X and Y respectively, divide the plane into rectangles such that each (real or imaginary) root lies alone by itself in a given rectangle; but the ulterior theory, even as regards the imaginary roots of an equation with real coefficients, has not been developed, and the remarks which immediately follow have reference only to equations with real coefficients, and to the real roots of such equations.

Supposing the roots separated as above, so that a certain root is known to lie alone by itself between two given limits, then it

is possible by various processes (Horner's, or Lagrange's method of continued fractions) to obtain to any degree of approximation the numerical value of the real root in question, and thus to obtain (approximately as above) the values of the several real roots.

The real roots can also frequently be obtained, without the necessity of a previous separation of the roots, by other processes of approximation—Newton's, as completed by Fourier, or by a method given by Encke—and the problem of their determination to any degree of approximation may be regarded as completely solved. But this is far from being practically the case even as regards the imaginary roots of such equations, or as regards the roots of an equation with imaginary coefficients.

A class of numerical equations which need to be considered, are the *binomial* equations $x^n - \alpha = 0$, where $\alpha = a + \beta i$, is a complex number. The foregoing conclusions apply, viz. there are always n roots, which it may be shown are all unequal. Supposing

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one of these is θ , so that $\theta^n = \alpha$, then, assuming $\theta = \alpha^{1/n}y$, we have $y^n - 1 = 0$, which equation (like the more general one $x^n - \alpha = 0$) has precisely n roots; it is shown that these are $1, \omega, \omega^2, \dots, \omega^{n-1}$, where ω is a complex number $a + \beta i$ such that $a^2 + \beta^2 = 1$, or, what is the same thing, a complex number of the form $\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$; and it then at once appears that ω may be taken $= e^{2\pi i/n}$. We have thus the trigonometrical solution of the equation $x^n - 1 = 0$. We may also obtain a like trigonometrical solution of the first-mentioned equation $x^n - \alpha = 0$. We are thus led to the notion (a numerical) of the radical $\alpha^{1/n}$, regarded as an n -valued function, viz. any one of these being denoted by $\sqrt[n]{\alpha}$, then the series of values is

$$\sqrt[n]{\alpha}, \omega \sqrt[n]{\alpha}, \dots, \omega^{n-1} \sqrt[n]{\alpha}.$$

Or we may, if we please, use $\sqrt[n]{\alpha}$, instead of $\alpha^{1/n}$, as a symbol to denote the n -valued function.

It is not necessary, as regards the equation $x^n - 1 = 0$, to refer here to the distinctions between the cases n a prime, and a composite, number.

As the coefficients of an algebraical equation may be numerical, all which follows in regard to algebraical equations, is (with, it may be, some few modifications) applicable to numerical equations; and hence, concluding for the present this subject, it will be convenient to pass on to algebraical equations.

II. We consider, secondly, an algebraical equation

$$x^n - p_1 x^{n-1} + \dots = 0,$$

and we here assume the existence of roots, viz. we assume that there are n quantities $a, b, c, \dots$ (in general, all of them different, but in particular cases they may become equal in sets in any manner), such that

$$x^n - p_1 x^{n-1} + \dots = (x - a)(x - b) \dots$$

Or, looking at the question in a different point of view, and starting with the roots $a, b, c, \dots$ as given, we express the product of the n factors $x - a, x - b, \dots$ in the foregoing form, and thus arrive at an equation of the order n having the n roots $a, b, c, \dots$. In either case, we have

$$p_1 = \Sigma a, \quad p_2 = \Sigma ab, \quad \dots, \quad p_n = abc \dots,$$

viz. regarding the coefficients $p_1, p_2, \dots, p_n$ as given, then we assume the existence of roots $a, b, c, \dots$ such that $p_1 = \Sigma a$, &c., or regarding the roots as given, then we write p_1, p_2 , &c., to denote the functions $\Sigma a, \Sigma ab$, &c.

It is to be noticed that, in virtue of

$$x^n - p_1 x^{n-1} + \dots = (x - a)(x - b) \dots, \quad \&c.,$$

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or of the equivalent equations $p_1 = \Sigma a$, &c., then

$$\begin{aligned} a^n - p_1 a^{n-1} + \dots &= 0, \\ b^n - p_1 b^{n-1} + \dots &= 0, \end{aligned}$$

(viz. it is for this reason that $a, b, \dots$ are said to be roots of $x^n - p_1 x^{n-1} + \dots = 0$); and, moreover, that conversely from the last-mentioned equations, assuming that $a, b, \dots$ are all different, we deduce

$$p_1 = \Sigma a, \quad p_2 = \Sigma ab, \quad \&c.,$$

and

$$x^n - p_1 x^{n-1} + \dots = (x - a)(x - b) \dots$$

Observe that, if for instance $a = b$, then the two equations $a^n - p_1 a^{n-1} + \dots = 0$, $b^n - p_1 b^{n-1} + \dots = 0$ would reduce themselves to a single equation, which would not of itself express that a was a double root, that is, that $(x - a)^2$ was a factor of $x^n - p_1 x^{n-1} + \dots$; but by considering b as the limit of $a + h$, h indefinitely small, we obtain a second equation

$$na^{n-1} - (n - 1)p_1 a^{n-2} + \dots = 0,$$

which, with the first, expresses that a is a double root; and then the whole system of equations leads, as before, to the equations $p_1 = \Sigma a$, &c. But this in passing: the general case is when the roots are all unequal.

We have then the theorem that every rational symmetrical function of the roots is a rational function of the coefficients; this is an easy consequence from the less general theorem, every rational and integral symmetrical function of the roots is a rational and integral function of the coefficients.

In particular, the sums of powers Σa^2 , Σa^3 , &c., are rational and integral functions of the coefficients.

An ordinary process, as regards the expression of other functions $\Sigma a^p b^q$, &c., in terms of the coefficients, is to make them

depend on the functions Σa^n , &c., but this is very objectionable; the true theory consists in showing that we have systems of equations

$$\begin{aligned} p_1 &= \Sigma a, \\ \left\{ \begin{aligned} p_2 &= \Sigma ab, \\ p_1^2 &= \Sigma a^2 + 2\Sigma ab, \end{aligned} \right. \\ \left\{ \begin{aligned} p_3 &= \Sigma abc, \\ p_1 p_2 &= \Sigma a^2 b + 3\Sigma abc, \\ p_1^3 &= \Sigma a^3 + 3\Sigma a^2 b + 6\Sigma abc, \end{aligned} \right. \\ &\quad \&c., \quad \&c., \end{aligned}$$

where, in each system, there are precisely as many equations as there are root-functions on the right-hand side, e.g. 3 equations and 3 functions Σabc , $\Sigma a^2 b$, Σa^3 . Hence, in each system, the root-functions can be determined linearly in terms of the powers and products of the coefficients.

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It follows that it is possible to determine an equation (of an assignable order) having for roots any given (unsymmetrical) functions of the roots of a given equation. For example, in the case of a quartic equation, roots (a, b, c, d) , it is possible to find an equation having the roots ab, ac, ad, bc, bd, cd , being therefore a sextic equation viz. in the product $(y - ab)(y - ac)(y - ad)(y - bc)(y - bd)(y - cd)$, the coefficients of the several powers of y will be symmetrical functions of a, b, c, d , and therefore rational and integral functions of the coefficients of the original quartic equation.

In connexion herewith, the question arises as to the number of values (obtained by permutations of the roots) of given *unsymmetrical* functions of the roots; for instance, with roots (a, b, c, d) as before, how many values are there of the function $ab + cd$; or, better, how many functions are there of this form; the answer is 3, viz. $ab + cd, ac + bd, ad + bc$; or, again, we may ask whether it is possible to obtain functions of a given number of values, 3-valued, 4-valued functions, &c.

We have, moreover, the very important theorem that, given the value of any unsymmetrical function, e.g. $ab + cd$, it is in general possible to determine rationally the value of any similar function, e.g. $(a + b)^2 + (c + d)^2$.

The *à priori* ground of this theorem may be illustrated by means of a numerical equation. Suppose, e.g. that the roots of a quartic equation are 1, 2, 3, 4; then if it is given that $ab + cd = 14$, this in effect determines a, b to be 1, 2 (viz. $a = 1, b = 2$, or else $a = 2, b = 1$) and c, d to be 3, 4 (viz. $c = 3, d = 4$, or else $c = 4, d = 3$); and it therefore in effect determines $(a + b)^2 + (c + d)^2$ to be = 370, and not any other value. And we can in the same way account for cases of failure as regards particular equations; thus, the roots being 1, 2, 3, 4, as above, $a - b = 2$ determines a to be = 1 and b to be = 2; but if the roots had been 1, 2, 4, 16, then $a^2b = 16$ does not uniquely determine a and b , but only makes them to be 1 and 16, or else 2 and 4, respectively.

As to the *à posteriori* proof, assume, for instance, $t_1 = ab + cd$, $y_1 = (a + b)^2 + (c + d)^2$, and so $t_2 = ac + db$, $y_2 = (a + c)^2 + (d + b)^2$,

&c.—in the present case there are only the functions t_1, t_2, t_3 and y_1, y_2, y_3 —then $y_1 + y_2 + y_3, t_1y_1 + t_2y_2 + t_3y_3, t_1^2y_1 + t_2^2y_2 + t_3^2y_3$ will be respectively symmetrical functions of the roots of the quartic, and therefore rational and integral functions of its coefficients, that is, they will be known.

Imagine, in the first instance, that t_1, t_2, t_3 are all known; then the equations being linear in y_1, y_2, y_3 , these can be expressed rationally in terms of known functions of the coefficients and of t_1, t_2, t_3 , that is, y_1, y_2, y_3 will be known. But observe further, that y_1 is obtained as a function of t_1, t_2, t_3 symmetrical as regards t_2, t_3 : it can consequently be expressed as a rational function of t_1 and of $t_2 + t_3, t_2t_3$, or, what is the same thing, of t_1 and $t_1 + t_2 + t_3, t_1t_2 + t_1t_3 + t_2t_3, t_1t_2t_3$; but these last will be symmetrical functions of the roots, and as such expressible rationally in terms of the coefficients: that is, y_1 will be expressed as a rational function of t_1 and of the coefficients, or, t_1 being known, y_1 will be rationally determined.

We may consider now the question of the algebraical solution of equations, or, more accurately, that of the solution of equations by radicals.

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In the case of a quadric equation $x^2 + px + q = 0$, we can find for x , by the assistance of the sign $\sqrt{(\)}$ or $(\)^{\frac{1}{2}}$ an expression for x as a two-valued function of the coefficients p, q , such that, substituting this value in the equation, the equation is thereby identically satisfied, viz. we have

$$x = -\frac{1}{2}p \pm \sqrt{(\frac{1}{4}p^2 - q)},$$

giving

$$\begin{array}{rcl} x^2 & = & \frac{1}{4}p^2 - q \mp p\sqrt{(\frac{1}{4}p^2 - q)} \\ +px & = & -\frac{1}{2}p^2 \pm p\sqrt{(\frac{1}{4}p^2 - q)} \\ +q & = & +q \\ \hline x^2 + px + q & = & 0, \end{array}$$

and the equation is on this account said to be algebraically solvable, or, more accurately, to be solvable by radicals. Or we may, by writing $x = -\frac{1}{2}p + z$, reduce the equation to $z^2 = \frac{1}{4}p^2 - q$, viz. to an equation of the form $z^2 = a$, and, in virtue of its being thus reducible, we may say that the equation is solvable by radicals. And the question for an equation of any higher order is, say of the order n , can we by means of radicals, that is, by aid of the sign $\sqrt[n]{(\)}$ or $(\)^{1/n}$, using as many as we please of such signs and with any values of m , find an n -valued function (or any function) of the coefficients, which substituted for x in the equation shall satisfy it identically.

It will be observed that the coefficients $p, q, \dots$ are not explicitly considered as numbers, but that even if they do denote numbers, the question whether a numerical equation admits of solution by radicals is wholly unconnected with the before-mentioned theorem of the existence of the n roots of such an equation. It does not even follow that, in the case of a numerical equation solvable by radicals, the algebraical expression of x gives the numerical solution; but this requires explanation. Consider, first, a numerical quadric equation with imaginary coefficients; in the formula $x = -\frac{1}{2}p \pm \sqrt{(\frac{1}{4}p^2 - q)}$, substituting for p, q their given numerical values we obtain for x an expression of the form $x = \alpha + \beta i \pm \sqrt{(\gamma + \delta i)}$, where $\alpha, \beta, \gamma, \delta$ are real numbers;

this value substituted in the numerical equation would satisfy it identically and it is thus an algebraical solution; but there is no obvious *à priori* reason why the expression $\sqrt{(\gamma + \delta i)}$ should have a value $= c + di$, where c and d are real numbers calculable by the extraction of a root or roots of real numbers; it appears upon investigation that $\sqrt{(\gamma + \delta i)}$ has such a value calculable by means of the radical expression $\sqrt{\{\sqrt{(\gamma^2 + \delta^2)} \pm \gamma\}}$; and hence that the algebraical solution of a quadric equation does in every case give the numerical solution of a numerical quadric. The case of a numerical cubic will be considered presently.

A cubic equation can be solved by radicals, viz. taking for greater simplicity the cubic in the reduced form $x^3 - qx - r = 0$, and writing $x = a + b$, this will be a solution if only $3ab = q$, and $a^3 + b^3 = r$, or say $\frac{1}{2}(a^3 + b^3) = \frac{1}{2}r$;

$$\frac{1}{2}(a^3 - b^3) = \pm \sqrt{(\frac{1}{4}r^2 - \frac{1}{27}q^3)},$$

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and therefore

$$a = \sqrt[3]{\left\{\frac{1}{2}r + \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}},$$

a six-valued function of q, r . But then writing $b = \frac{q}{3a}$, we have, as may be shown, $a + b$ a three-valued function of the coefficients; it would have been wrong to complete the solution by writing $b = \sqrt[3]{\left\{\frac{1}{2}r \pm \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}}$, since here $(a + b)$ would be given as a 9-valued function, having only 3 of its values roots, and the other 6 values being irrelevant. An interesting variation of the solution is to write $x = ab(a + b)$, giving $a^3b^3(a^3 + b^3) = r$ and $3a^2b^2 = q$, or say $\frac{1}{2}(a^3 + b^3) = \frac{r}{2}$, $a^3b^3 = \frac{q^3}{27}$; whence

$$\left\{\frac{1}{2}(a^3 - b^3)\right\}^2 = \frac{1}{4}r^2 - \frac{1}{27}q^3,$$

and therefore

$$a = \sqrt[3]{\left\{\frac{1}{2}r + \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}}, \quad b = \sqrt[3]{\left\{\frac{1}{2}r - \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}},$$

and here although a, b are each of them a 6-valued function, yet, as may be shown, $ab(a + b)$ is only a 3-valued function.

In the case of a numerical cubic, even when the coefficients are real, substituting their values in the expression

$$x = \sqrt[3]{\left\{\frac{1}{2}r + \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}} + \sqrt[3]{\left\{\frac{1}{2}r - \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}},$$

this may depend on an expression of the form $\sqrt[3]{(\gamma + \delta i)}$, where γ and δ are real numbers (viz. it will do so if $\frac{1}{4}r^2 - \frac{1}{27}q^3$ is a negative number), and here we cannot by the extraction of any root or roots of real numbers reduce $\sqrt[3]{(\gamma + \delta i)}$ to the form $c + di$, c and d real numbers; hence, here the algebraical solution does not give the numerical solution. It is to be added that the case in question, called the “irreducible case,” is that wherein the three roots of the cubic equation are all real; if the roots are one real and two imaginary, then, contrariwise, the quantity

under the cube root is real, and the algebraical solution gives the numerical one.

The irreducible case is solvable by a trigonometrical formula, but this is not a solution by radicals; it consists, in effect, in reducing the given numerical cubic (not to a cubic of the form $z^3 = a$, solvable by the extraction of a cube root, but) to a cubic of the form $4x^3 - 3x = a$, corresponding to the equation $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ which serves to determine $\cos\theta$ when $\cos 3\theta$ is known.

A quartic equation is solvable by radicals; and it may be remarked, that the existence of such a solution depends on the existence of 3-valued functions such as $ab + cd$, of the four roots (a, b, c, d) ; by what precedes, $ab + cd$ is the root of a cubic equation, which equation is solvable by radicals; hence $ab + cd$ can be found by radicals; and since $abcd$ is a given value, ab and cd can each be found by radicals. But by what precedes, if ab be known, then any similar function, say $a + b$, is obtainable rationally; and, consequently, from the values of $a + b$ and ab we may by radicals

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obtain the value of a or b , that is, an expression for a root of the given quartic expression; the expression finally obtained is 4-valued, corresponding to the different values of the several radicals which enter therein, and we have therefore the expression by radicals of each of the four roots of the quartic equation. But when the quartic is numerical, the same thing arises as in the cubic: the algebraical expression does not in every case give the numerical one.

It will be understood from the foregoing explanation as to the quartic, how in the next following case, that of a quintic equation, the question of the solvability by radicals depends on the existence or non-existence of r -valued functions of the five roots (a, b, c, d, e) ; a fundamental theorem on the subject is that a rational function of 5 letters, if it has less than 5, cannot have more than 2 values; viz. that there are no 3-valued, or 4-valued, functions of 5 letters; and by reasoning, depending in part upon this theorem, Abel showed that a general quintic equation is not solvable by radicals: and *à fortiori* the general equation of any order higher than 5 is not solvable by radicals.

The general theory of the solvability of an equation by radicals depends very much on Vandermonde's remark, that supposing an equation is solvable (by radicals) and that we have therefore an algebraical expression of x in terms of the coefficients, then substituting for the coefficients their values in terms of the roots, the resulting value of the expression must reduce itself to any one at pleasure of the roots $a, b, c, \dots$; thus in the case of the quadric equation where the solution is $x = -\frac{1}{2}p \pm \sqrt{(\frac{1}{4}p^2 - q)}$, writing for p, q their values $a + b, ab$, this is $x = \frac{1}{2}(a + b) \pm \sqrt{\{(a - b)^2\}}$, $= a$ or b according to the value of the radical. But it is not considered necessary in the present sketch to go further into the theory of the solvability of an equation by radicals. It may be proper to remark that, for quintic equations, there are solutions analogous to the trigonometrical solution of a cubic equation, viz. the quintic equation is here in effect reduced to some special form of quintic equation; for instance, to Jerrard's form $x^5 + ax + b = 0$ or to some form presenting itself in the

theory of elliptic functions; but the solutions in question are not solutions by radicals. And there are various other interesting parts of the theory which have been excluded from consideration.

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ON ARONHOLD'S INTEGRATION-FORMULA.

[From the *Messenger of Mathematics*, vol. v. (1876),
pp. 88—90.]

The fundamental theorem in Aronhold's Memoir, "Ueber eine neue algebraische Behandlungsweise der Integrale $\int \Pi(x, y) dx$, &c.," *Crelle*, t. LXI. (1863), pp. 95—145, is a theorem of indefinite integration. The form is

$$\Lambda \int \frac{dx}{(\alpha x + \beta y + \gamma)(hx + by + f)} = \log \frac{(a\xi + h\eta + g)x + (h\xi + b\eta + f)y + g\xi}{\alpha x + \beta y + \gamma}$$

where y is a certain irrational function of x , determined by a quadric equation, and the other symbols denote constants connected by certain relations; viz. writing, for shortness,

$$U = (a, b, c, f, g, h \parallel x, y, 1)^2 = (a, \dots \parallel x, y, 1)^2, \text{ for shortness,}$$

that is,

$$U = ax^2 + 2hxy + by^2 + 2fy + 2gx + c;$$

$$W = (a, b, c, f, g, h \parallel x, y, 1 \parallel \xi, \eta, 1) = (a, \dots \parallel x, y, 1 \parallel \xi, \eta, 1),$$

that is,

$$W = (ax + hy + g)\xi + (hx + by + f)\eta + gx + fy + c,$$

or

$$= (a\xi + h\eta + g)x + (h\xi + b\eta + f)y + g\xi + f\eta + c;$$

$$(P, Q, R) = (ax + hy + g, hx + by + f, gx + fy + c),$$

$$(P_0, Q_0, R_0) = (a\xi + h\eta + g, h\xi + b\eta + f, g\xi + f\eta + c),$$

$$\Omega = \alpha x + \beta y + \gamma,$$

$$\Omega_0 = \alpha \xi + \beta \eta + \gamma,$$

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

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then y is determined as a function of x by the equation $U = 0$, that is,

$$(a, b, c, f, g, h \setminus x, y, 1)^2 = 0;$$

or, what is the same thing,

$$by = -(hx + f) + \sqrt{\{-Cx^2 + 2Gx - A\}};$$

the constants $\alpha, \beta, \gamma, \xi, \eta$ are such that

$$(a, b, c, f, g, h \setminus \xi, \eta, 1)^2 = 0, \quad \alpha\xi + \beta\eta + \gamma = 0,$$

that is,

$$U_0 = 0, \quad \Omega_0 = 0;$$

and the value of Λ is given by

$$\Lambda^2 = -(A, B, C, F, G, H \setminus \alpha, \beta, \gamma)^2.$$

The theorem may therefore be written

$$\Lambda \int \frac{dx}{\Omega \cdot Q} = \log \frac{W}{\Omega},$$

where the several symbols have the significations explained above.

The verification is as follows. We ought to have

$$\frac{\Lambda dx}{\Omega Q} = \frac{P_0 dx + Q_0 dy}{W} - \frac{\alpha dx + \beta dy}{\Omega},$$

when dx, dy satisfy the relation $P dx + Q dy = 0$, viz. substituting for dy the value $-\frac{P}{Q} dx$, the equation becomes

$$\frac{\Lambda}{\Omega \cdot Q} = \frac{P_0 Q - P Q_0}{W \cdot Q} - \frac{\alpha Q - \beta P}{\Omega \cdot Q},$$

that is, substituting for Ω its value,

$$\Lambda W = (P_0 Q - P Q_0)(\alpha x + \beta y + \gamma) - (\alpha Q - \beta P) W.$$

On the right-hand side, substituting for W its value,

$$\text{coeff. } \alpha = x(P_0Q - PQ_0) - Q(P_0x + Q_0y + R_0), \quad = Q_0R - QR_0,$$

$$\text{coeff. } \beta = y(P_0Q - PQ_0) + P(P_0x + Q_0y + R_0), \quad = R_0P - RP_0,$$

(as at once appears by aid of the relation $U = Px + Qy + R = 0$),

$$\text{coeff. } \gamma = P_0Q - PQ_0.$$

The equation to be verified thus is

$$\Lambda W = \begin{vmatrix} \alpha & \beta & \gamma \\ P_0 & Q_0 & R_0 \\ P & Q & R \end{vmatrix},$$

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which, substituting therein for P, Q, R, P_0, Q_0, R_0 , their values, and writing

$$(\lambda, \mu, \nu) = (\eta - y, x - \xi, \xi y - \eta x),$$

is in fact

$$\Lambda W = (A, \dots \backslash \lambda, \mu, \nu \backslash \alpha, \beta, \gamma).$$

We have identically

$$(a, \dots \backslash x, y, 1)^2 (a, \dots \backslash \xi, \eta, 1)^2 - W^2 = (A, \dots \backslash \lambda, \mu, \nu)^2,$$

which, in virtue of $(a, \dots \backslash \xi, \eta, 1)^2 = 0$, gives

$$W^2 = -(A, \dots \backslash \lambda, \mu, \nu)^2,$$

and since $\Lambda^2 = -(A, \dots \backslash \alpha, \beta, \gamma)^2$, the equation is thus

$$\sqrt{\{-(A, \dots \backslash \alpha, \beta, \gamma)^2\}} \sqrt{\{-(A, \dots \backslash \lambda, \mu, \nu)^2\}} = (A, \dots \backslash \lambda, \mu, \nu \backslash \alpha, \beta, \gamma),$$

that is,

$$(A, \dots \backslash \alpha, \beta, \gamma)^2 (A, \dots \backslash \lambda, \mu, \nu)^2 - \{(A, \dots \backslash \lambda, \mu, \nu \backslash \alpha, \beta, \gamma)\}^2 = 0.$$

The left-hand side is here identically

$$= K (\alpha\mu - \beta\lambda, \alpha\nu - \gamma\lambda, \beta\nu - \gamma\mu)^2,$$

substituting for λ, μ, ν their values, we find

$$(\alpha\mu - \beta\lambda, \alpha\nu - \gamma\lambda, \beta\nu - \gamma\mu) = (\xi\alpha + \eta\beta + \gamma)(\alpha, \beta, \gamma) - \Omega(\xi, \eta, 1),$$

viz. in virtue of $\Omega_0 = 0$, these are $= -\xi\Omega, -\eta\Omega, -\Omega$, and the quadric function is $= \Omega^2(a, \dots \backslash \xi, \eta, 1)^2$, vanishing in virtue of the relation $(a, \dots \backslash \xi, \eta, 1)^2 = 0$.

The equation in question

$$\sqrt{\{-(A, \dots \backslash \alpha, \beta, \gamma)^2\}} \sqrt{\{-(A, \dots \backslash \lambda, \mu, \nu)^2\}} = (A, \dots \backslash \lambda, \mu, \nu \backslash \alpha, \beta, \gamma)$$

is thus verified, and the theorem is proved.

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**NOTE ON MR. MARTIN'S PAPER, "ON THE
INTEGRALS OF SOME DIFFERENTIALS."**

[From the *Messenger of Mathematics*, vol. v. (1876), p. 163.]

The Note refers to a detail in a process of integration.

634.

THEOREMS IN TRIGONOMETRY AND ON
PARTITIONS.

[From the *Messenger of Mathematics*, vol. v. (1876), p. 164,
and p. 188.]

If

$$A + B + C + F + G + H = 0,$$

then

$$\begin{vmatrix} \sin \overline{A + F} & \sin \overline{B + F} & \sin \overline{C + F} & \cos F & \sin F \\ \sin \overline{A + G} & \sin \overline{B + G} & \sin \overline{C + G} & \cos G & \sin G \\ \sin \overline{A + H} & \sin \overline{B + H} & \sin \overline{C + H} & \cos H & \sin H \end{vmatrix} = 0.$$

(The determinant is to be read with the convention that, for instance, $\sin \overline{A + F}$ denotes $\sin(A + F)$.)

Let u_n = number of partitions of n , no part less than 2, the order attended to; e.g. if $n = 7$, the partitions are 7, 52, 25, 43, 34, 322, 232, 223, $u_7 = 8$; the series is

$$u_2 = 1,$$

$$u_3 = 1,$$

$$u_4 = 2,$$

$$u_5 = 3,$$

$$u_6 = 5,$$

$$u_7 = 8,$$

$$u_8 = 13,$$

$$u_9 = 21,$$

where each term is the sum of the next preceding two terms.

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NOTE ON THE DEMONSTRATION OF CLAIRAUT'S THEOREM.

[From the *Messenger of Mathematics*, vol. v. (1876), pp. 166,
167.]

It seems worth while to indicate what the leading steps of the demonstration are.

The potential of the Earth's mass upon an external or superficial point is taken to be

$$V = \frac{V_0}{r} + \frac{V_1}{r^2} + \frac{V_2}{r^3} + \&c.,$$

where $V_1, V_2, V_3, \dots$ are Laplace's functions of the angular coordinates.

The surface is assumed to be a nearly spherical surface $r = a(1 + u)$, where $u = u_1 + u_2 + \&c.$, and $u_1, u_2, \dots$ are Laplace's functions of the angular coordinates. To be a surface of equilibrium, with an equation $V + \frac{1}{2}\omega^2 r^2 \sin^2 \theta = C$, the latter must be equivalent to the equation $r = a(1 + u)$, and it follows that we have

$$\begin{aligned} V_1 &= V_0 a u_1, \\ V_2 &= V_0 a^2 u_2 - \frac{1}{2}\omega^2 a^4 \left(\frac{1}{3} - \cos^2 \theta\right), \\ &\&c., \end{aligned}$$

which values are to be substituted in the expression for V .

The whole force of gravity (due to the attraction and the centrifugal force) is taken to be $g = -\frac{d}{dr}(V + \frac{1}{2}\omega^2 r^2 \sin^2 \theta)$, and it follows that

$$g = \frac{V_0}{a^2}(1 + u_1 + 2u_2 + \dots) - \frac{2}{3}\omega^2 a - \omega^2 a \left(\frac{1}{3} - \cos^2 \theta\right),$$

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which is of the form

$$g = G \left\{ 1 + u_1 - \frac{\omega^2 a}{G} \left(\frac{1}{3} - \cos^2 \theta \right) + 2u_2 + \dots \right\}.$$

Taking the Earth to be the spheroid of revolution

$$r = a \{ 1 + \epsilon \left(\frac{1}{3} - \cos^2 \theta \right) \},$$

then

$$u_1 = \epsilon \left(\frac{1}{3} - \cos^2 \theta \right), \quad u_2 = 0, \quad \&c.,$$

and the equation is

$$g = G \left\{ 1 - \left(\frac{\omega^2 a}{G} - \epsilon \right) \left(\frac{1}{3} - \cos^2 \theta \right) \right\},$$

or say

$$g = G \{ 1 - \left(\frac{5}{2} m - \epsilon \right) \left(\frac{1}{3} - \cos^2 \theta \right) \},$$

where $m = \frac{\omega^2 a}{G}$, is the ratio of the centrifugal force at the equator to the force of gravity, which is the theorem in question. The expression “it follows” has been twice used as meaning it follows as a mere analytical consequence, in the proper degree of approximation, the steps of the deduction being purposely omitted.

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ON THE THEORY OF THE SINGULAR
SOLUTIONS OF DIFFERENTIAL EQUATIONS OF
THE FIRST ORDER.

[From the *Messenger of Mathematics*, vol. vi. (1877),
pp. 23—27.]

In continuation of the former paper with this title (*Messenger*, vol. ii., 1873, pp. 6—12, [545]), I propose to discuss various particular examples, chiefly of cases in which the differential equation is of the form $(L, M, N \setminus\setminus p, 1)^2 = 0$, where L, M, N are rational and integral functions of (x, y) , and whether it admits or does not admit of an integral equation $(P, Q, R \setminus\setminus c, 1)^2 = 0$, where P, Q, R are rational and integral functions of (x, y) .

The singular solution of the differential equation

$$(L, M, N \setminus\setminus p, 1)^2 = 0,$$

if there be a singular solution, is $S = 0$, where S is either $= LN - M^2$, or a factor of $LN - M^2$. But in general $LN - M^2$ is an indecomposable function, such that $LN - M^2 = 0$ is not a solution of the differential equation, and this being so, there is no singular solution; viz. a differential equation $(L, M, N \setminus\setminus p, 1)^2 = 0$, where L, M, N are rational and integral functions of (x, y) , has not in general any singular solution.

Consider now a system of algebraical curves $U = 0$, where U is as regards (x, y) a rational and integral function of the order m , and depends in any manner on an arbitrary parameter c^1 . I say that there is always a proper envelope, which envelope is the singular solution of the differential equation obtained by the elimination of c from the equation $U = 0$, and the derived equation in regard to (x, y) . It follows that the differential equation $(L, M, N \setminus\setminus p, 1)^2 = 0$, which has no singular solution, does

¹The expressions in the text may be understood as extending to the case where U is a function of any number (o) of constants $c_1, c_2, \dots, c_o$, connected by an $(o - 1)$ -fold relation, U thus virtually depending on a single arbitrary parameter.

not admit of an integral of the form in question $U = 0$, viz. an integral representing a system of algebraic curves.

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The theorem just referred to, that the system of algebraic curves $U = 0$ has always an envelope, is an interesting theorem, which I proceed to prove. Assume that in general, that is, for an arbitrary value of the parameter, the equation $U = 0$ represents a curve of the order m , with δ nodes and κ cusps (and therefore of the class n , with ι inflexions and τ double tangents, the numbers $m, \delta, \kappa, n, \tau, \iota$ being connected by Plücker's equations); for particular values of the parameter, the values of δ and κ may be increased, or the curve may break up, but this is immaterial.

The consecutive curve $U + \delta c \cdot d_c U = 0$ is a curve of the same order m , with δ nodes and κ cusps, consecutive to the nodes and cusps of the original curve U , and the two curves intersect in m^2 points; but of these, there are 2 coinciding with each node, and 3 coinciding with each cusp of the curve $U = 0$, as at once appears by drawing a curve with a node or a cusp, and the consecutive curve with a consecutive node or cusp; the number of the remaining intersections is $= m^2 - 2\delta - 3\kappa$, and the envelope is the locus of these $m^2 - 2\delta - 3\kappa$ points. Observe that the two curves have in common n^2 tangents; but of these, 2 coincide with each double tangent and 3 coincide with each stationary tangent of the curve $U = 0$, viz. the number of the remaining common tangents is $= n^2 - 2\tau - 3\iota$ (which is $= m^2 - 2\delta - 3\kappa$): and that these $n^2 - 2\tau - 3\iota$ common tangents are indefinitely near to the $m^2 - 2\delta - 3\kappa$ common points respectively, and are in fact the tangents of the envelope at the $m^2 - 2\delta - 3\kappa$ points respectively. Now in an algebraic curve we have $m + n = m^2 - 2\delta - 3\kappa$, viz. the number $m^2 - 2\delta - 3\kappa$ cannot be $= 0$, and we have therefore always an envelope the locus of the system of the $m^2 - 2\delta - 3\kappa$ points. It might be thought that the conclusion extends to transcendental curves; if this were so, the result would prove too much, viz. it would follow that a differential equation $(L, M, N \setminus p, 1)^2 = 0$ without a singular solution had no general integral; but it will appear by an example that the theorem as to the envelope does not extend to transcendental curves.

Ex. 1.

$$p^2 - (1 - y^2) = 0, \text{ that is, } dy^2 - (1 - y^2) dx^2 = 0.$$

Here there is no algebraical integral, but there is a quasi-algebraical integral of the form $(P, Q, R \setminus c, 1)^2 = 0$; viz. starting with the form $y = \sin(x + c)$ and expressing $\sin c$ and $\cos c$ rationally in terms of a new parameter, this is

$$c^2(y + \cos x) - 2c \sin x + (y - \cos x) = 0,$$

where the coefficients are one-valued functions of (x, y) . The discriminant of the differential equation in regard to p and that of the integral equation in regard to c are each $= y^2 - 1$, and we have a true singular solution $y^2 - 1 = 0$.

Ex. 2.

$$(1 - x^2)p^2 - (1 - y^2) = 0,$$

that is,

$$(1 - x^2) dy^2 - (1 - y^2) dx^2 = 0.$$

We have here an algebraic integral of the proper form, which is at once derived from the circular form

$$C = \cos^{-1} x + \cos^{-1} y$$

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by changing the constant, viz. this is

$$c^2 - 2cxy - (1 - x^2 - y^2) = 0.$$

The two discriminants are here each $= (x^2 - 1)(y^2 - 1)$, and we have

$$(x^2 - 1)(y^2 - 1) = 0$$

as a true singular solution. The curves are in fact the system of conics (ellipses and hyperbolas) each touching the four lines $x = 1, x = -1, y = 1, y = -1$.

Ex. 3.

$$(1 - y^2)p^2 - 1 = 0, \quad \text{that is,} \quad (1 - y^2)dy^2 - dx^2 = 0.$$

This is an extremely interesting example: the curve is the orthogonal trajectory of the system of sinusoids $y = \sin(x + c)$, which is the integral of Example 1; and we thus at once see that the real portion of the curve is wholly included between the lines $y = -1, y = +1$, being an infinite continuous curve, having a series of equidistant cusps alternately at the one and the other line, and obtained by the continued repetition of the finite portion included between two consecutive cusps on the same line. The discriminant of the differential equation equated to zero gives $y^2 - 1 = 0$, the equation of the two lines in question; but this does not satisfy the differential equation, and it is consequently not a singular solution; by what precedes, it appears that it is, in fact, a cusp-locus.

We thus see that the curves which represent the integral equation have no real envelope; but it is to be further shown that there is no imaginary envelope, and that the curve obtained by the elimination of the parameter is, in fact, made up of a (imaginary) node-locus and of the foregoing cusp-locus.

The curve is properly represented by taking x, y each of them a one-valued function of the parameter θ , viz. we may write

$$\begin{aligned} y &= \cos \theta, \\ x &= c + \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta. \end{aligned}$$

In fact, these values give

$$\frac{dy}{d\theta} = -\sin \theta, \quad \frac{dx}{d\theta} = \frac{1}{2}(1 - \cos 2\theta) = \sin^2 \theta,$$

and therefore

$$p = \frac{-1}{\sin \theta} = \frac{-1}{\sqrt{(1 - y^2)}},$$

that is, $(1 - y^2)p^2 - 1 = 0$, the differential equation.

It is obvious that to a given value of the parameter there corresponds a single point of the curve; and it is to be shown that, conversely, to a given point of the curve corresponds in general a single value of the parameter.

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Suppose the coordinates of the given point are $y = \cos \alpha$, $x = c + \frac{1}{2}\alpha - \frac{1}{4}\sin 2\alpha$, where α is a determinate quantity; then, to find θ , we have

$$\cos \theta = \cos \alpha, \quad 2\theta - \sin 2\theta = 2\alpha - \sin 2\alpha.$$

The first equation gives $\theta = 2m\pi \pm \alpha$, and the second equation then is

$$4m\pi \pm 2\alpha \mp \sin 2\alpha = 2\alpha - \sin 2\alpha,$$

viz. taking the upper signs, this is $4m\pi = 0$, giving $m = 0$ and $\theta = \alpha$; and, taking the lower signs, it is $m\pi = \alpha - \sin \alpha$, which, α being given, is not in general satisfied; hence to the given point there corresponds only the value α of the parameter θ . If, however, α is such that $\alpha - \sin \alpha$ is equal to a multiple of π , say $r\pi$, then the last-mentioned equation is satisfied by the value $m = r$, so that to the given point of the curve correspond the two values α and $2r\pi - \alpha$ of the parameter; these values are in general unequal, and the point is then a node; but they may be equal, viz. this is so if $\alpha = r\pi$ (the point on the curve being then $y = \cos r\pi, = \pm 1, x = c + \frac{1}{2}r\pi$), and the point is then a cusp; showing what was known, that there are on each of the lines $y = -1, y = +1$, an infinite series of equidistant cusps.

More definitely, suppose $\alpha = r\pi \pm \beta$, where β is a root of the equation $2\beta - \sin 2\beta = 0$, then

$$\sin 2\alpha = \pm \sin 2\beta, \quad 2\alpha - \sin 2\alpha = 2r\pi + (2\beta - \sin 2\beta) = 2r\pi,$$

and to the given point on the curve correspond the two values α and $2r\pi - \alpha$ of the parameter. If $\beta = 0$, we have, as above, the cusps on the two lines $y = +1, y = -1$ respectively; but if β be an imaginary root of the equation $2\beta - \sin 2\beta = 0$, then we have an infinite series of nodes on the imaginary line $y = \cos r\pi \cos \beta$; and there are an infinite number of such lines corresponding to the different imaginary roots of the equation $2\beta - \sin 2\beta = 0$.

From the form in which the equation of the curve is given, we cannot directly form the equation of the envelope by equating

to zero the discriminant in regard to the constant c ; but we may determine the intersections of the curve by the consecutive curve (corresponding to a value $c + \delta c$ of the constant), and thus determine the locus of these intersections.

Consider for a moment the curves belonging to the constants c , c_1 , and let θ, θ_1 be the values of the parameter θ belonging to the points of intersection; we have $\cos \theta = \cos \theta_1$, $4c + 2\theta - \sin 2\theta = 4c_1 + 2\theta_1 - \sin 2\theta_1$, we have $\theta_1 = 2r\pi + \theta$, but we cannot thereby satisfy the second equation; or else $\theta_1 = 2r\pi - \theta$, giving

$$4c + 2\theta - \sin 2\theta = 4c_1 + 4r\pi - 2\theta + \sin 2\theta,$$

that is, $2\theta - \sin 2\theta = 2c_1 - 2c + 2r\pi$; and we have thus corresponding to any given value of r a series of values of θ , viz. these are $\theta = r\pi + \beta$, where β is any root of the equation

$$2\beta - \sin 2\beta = 2c_1 - 2c.$$

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In particular, taking $c_1 = c$, the intersections are given by $\theta = r\pi + \beta$, where β is any root of the equation $2\beta - \sin 2\beta = 0$; viz. we have thus an infinite number of intersections lying on each of the lines $y = \cos r\pi \cos \beta$. If $\beta = 0$, the intersections lie on the two lines $y = 1, y = -1$ respectively; if β be an imaginary root of the equation $2\beta - \sin 2\beta = 0$, then they lie on the imaginary lines $y = \cos r\pi \cos \beta$. But by what precedes, it is clear that in the former case the intersections are nothing else than the cusps on the lines $y = 1, y = -1$; and in the latter case nothing else than the nodes on the lines $y = \cos r\pi \cos \beta$; viz. there is no proper envelope, but instead thereof we have lines of cusps and of nodes.

Ex. 4.

$$(1 - y^2)p^2 - (1 - x^2) = 0,$$

that is,

$$(1 - y^2) dy^2 - (1 - x^2) dx^2 = 0.$$

I have not examined this; the curve is the series of orthogonal trajectories of the conics of Example 2, and the integral equation may be represented by $y = \cos \theta, x = \cos \phi$, where $c = (2\theta - \sin 2\theta) - (2\phi - \sin 2\phi)$.

Equating to zero the discriminant of the differential equation, we have $(1 - y^2)(1 - x^2) = 0$, viz. the four lines $x = 1, x = -1, y = 1, y = -1$; this is not an envelope, but a locus of cusps.

637.

ON A DIFFERENTIAL EQUATION IN THE
THEORY OF ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. vi. (1877), p. 29.]

In the differential equation

$$Q^2 - Q\left(k + \frac{1}{k}\right) - 3 = 3(1 - k^2)\frac{dQ}{dk},$$

considered *Messenger*, t. iv., pp. 69 and 110, [594] and [597], writing $Q = x$ and $k + \frac{1}{k} = y$, the equation becomes

$$dy = \frac{3(y^2 - 4)dx}{3 + xy - x^3},$$

and we have, as a particular solution,

$$y = \frac{1}{4}\left(x^2 - 6x - \frac{3}{x}\right).$$

To verify this, observe that from the value of y

$$dy = \frac{3}{4x^2}(x^3 - 1)^2 dx, \quad 3 + xy - x^3 = \frac{1}{4}(x^3 - 1)(x^3 - 9),$$

and the equation becomes

$$\frac{3}{4x^2}(x^3 - 1)^2 = \frac{3}{16x^2} \frac{\{(x^3 - 6x^2 - 3)^2 - 64x^3\}}{\frac{1}{4}(x^3 - 1)(x^3 - 9)},$$

viz. this is

$$(x^3 - 1)^2(x^3 - 9) = (x^3 - 6x^2 - 3)^2 - 64x^3,$$

which is right.

638.

ON A q -FORMULA LEADING TO AN EXPRESSION FOR E_1 .

[From the *Messenger of Mathematics*, vol. vi. (1877),
pp. 63—66.]

It is to be shown that we have identically

$$\begin{aligned} (1 + 2q + 2q^4 + \dots)^4 - 16 \left(\frac{q}{1 - q^2} + \frac{2q^2}{1 - q^4} + \frac{3q^3}{1 - q^6} + \dots \right) \\ = \frac{1 - 9q - 25q^2 + 49q^4 + 81q^6 - \dots}{1 - q - q^2 + q^5 + q^7 - \dots} \quad (A); \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} (1 - 2q + 2q^4 - 2q^9 + \dots)^4 - 16 \left(\frac{-q}{1 - q^2} + \frac{2q^2}{1 - q^4} - \frac{3q^3}{1 - q^6} + \dots \right) \\ = \frac{1 + 9q + 25q^2 + 49q^4 + 81q^6 + \dots}{1 + q + q^2 + q^4 + q^6 + \dots} \quad (B), \end{aligned}$$

where the form (A) is that intended to be made use of, but the form (B) is rather more convenient for the demonstration.

We have

$$(1 - 2q + 2q^4 - 2q^9 + \dots)^4 = 1 + 8 \left\{ \frac{-q}{1 + q} + \frac{2q^2}{1 + q^2} - \frac{3q^3}{1 + q^3} + \dots \right\},$$

(Jacobi, *Fund. Nova*, p. 188, *Ges. Werke*, t. i., p. 239), taking the formula as there written down, and changing q into $-q$.

Also, if for a moment

$$X = 1 + q + q^2 + q^4 + q^6 + \&c.,$$

and

$$X' = \frac{dX}{dq},$$